Noakhali Science and Technology University
Department of Computer Science & Telecommunication Engineering

1st Year 1st Term M.Sc. (Engg.) Session: 2022-23 Final Examination- May, 2025
Course Code: CSTE 5131 Course Title: Machine Learning
Time: 4 Hours Total Marks: 70

[Answer any seven of the following questions. Figures in the right hand margin indicate full marks]

1. a) How does the quality and dimensionality of feature vectors affect model performance? 2
b) Suppose you train a decision tree classifier with a depth of 10 and get high training accuracy but low test accuracy. How would this affect generalization, and what adjustments could you make to improve it? 3
c) Compare linear and non-linear decision boundaries. When might a linear boundary be insufficient? 3
d) How can you ensure that both training and testing sets maintain the same class distribution when splitting a dataset? What method supports this? 2

2. a) Suppose we train a model to predict whether a patient has a certain disease (Positive) or Not (Negative). After training the model, we apply it to a test set of 1000 new patient records (each with a true label), and the model produces the following confusion matrix. 3

		Actual Class	
		Positive	Negative
Predicted class	Positive	220	120
	Negative	60	600

Compute the **F-score** of this model with respect to the Positive class.
b) Why is a p-value used in hypothesis testing, and what does a p-value < 0.05 indicate? 2
c) What considerations should be taken into account when choosing the appropriate number of folds for cross-validation? 2
d) What is oversampling and undersampling in the context of imbalanced datasets, and why are they used in machine learning? 3

3. a) You are given a dataset concerning whether a customer is likely to be approved for a loan based on some basic attributes. Each row in the dataset represents a customer, and the class attribute is the **loan approval status**: Approved or Denied. Attributes are 6

Customer Id: Unique identifier
Income level: High, Medium, Low
Credit score: Good, Bad
Employment status: Employed, Unemployed
Loan approval: Approved (A), Denied (D) ← (Target class)

Customer ID	Income level	Credit score	Employment status	Loan approval
1	High	Good	Employed	A
2	Low	Bad	Unemployed	D
3	Medium	Good	Employed	A
4	Low	Good	Employed	D
5	High	Bad	Unemployed	D
6	Medium	Bad	Employed	A

Which attribute will be chosen by the Information Gain criterion as the decision tree root and why? Show the detail calculation.

b) A decision tree splits data using the following conditions: $X1 \leq 4$ 4
$X2 > 3$
$X1 \leq 6$
Draw the axis-parallel decision boundaries in the X1-X2 plane and label each region with its class.

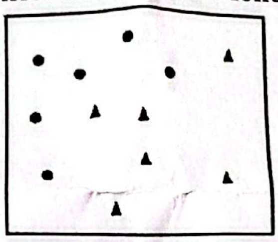
4. a) The following table contains 8 entities with color, legs, height, smelly, and species. 6

Sl. No.	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

We want to classify a new entity with Color=Green, Legs=2, Height=Tall, Smelly=No. Use naïve-bayes algorithm and tell the species of this test instance.

b) What is the difference between K-means and hierarchical clustering? Explain the concept of centroid linkage in hierarchical clustering. 4

5. a) Consider training AdaBoost using decision stumps on the following dataset. 4



How many iterations do you think it will take to achieve zero training error? Explain.

b) Explain the concept of Ensemble learning in the context of perceptron model through the expansion of the hypothesis space. 3

c) How does Ensemble learning apply in Random Forests to improve model performance? 3

6. a) Consider the following and justify your answer: 6

(i) The error rate of the ensemble classifier never increases from one round to the next. True or False?

(ii) Adaboost accounts for outliers by lowering the weights of training points that are repeatedly misclassified. True or False?

(iii) When you update weights, the training point with the smallest weight in the previous round will always increase in weight. True or False?

b) Discuss about support vector machine (SVM). What is the basic principle of SVM? Explain. 4

7. a) What do you mean by K-means clustering? How do you choose the number of clusters in k-means? Explain. 3

b) For the given data, compute two clusters using K-means algorithm for clustering where initial cluster centers are (1.0, 1.0) and (5.0, 7.0). Execute for two iterations. 7

Record Number	A	B
R1	1.0	1.0
R2	1.5	2.0
R3	3.0	4.0
R4	5.0	7.0
R5	3.5	5.0
R6	4.5	5.0
R7	3.5	4.5

8\. a) You are given the following distance matrix between five data points A, B, C, D, and E: 3

	A	B	C	D	E
A	0				
B	4	0			
C	2	5	0		
D	6	3	6	0	
E	7	8	9	4	0

Perform agglomerative hierarchical clustering and draw the dendrogram representing the clustering process using the single linkage method.

b) Suppose the data mining task is to cluster points into two clusters (K = 2), where the points (or data objects) are A (1,2), B (2,1), C (4,3), D (5,4). 4

The distance function is Euclidean distance. Initially we assign A and B as the center of each cluster, respectively. Use the *k-means* algorithm to show the final two clusters including cluster centers and cluster members.

c) Consider the following dataset of four individuals described by three nominal attributes: 3

Person	Color Preference	Vehicle Type	Subscription
A	Red	Car	Yes
B	Blue	Bike	No
C	Red	Car	No
D	Green	Car	Yes

Compute the dissimilarity matrix between all pairs of individuals using the simple matching method.

9\. a) A 28 × 28 image passes through the following layers in sequence: 3

- Convolution layer: 16 filters, 3×3 kernel, padding=1, stride = 1
- Max Pooling: 2×2, no padding, stride = 2

What is the output shape?

b) How do advancements in CNN architectures, such as residual connections, attention mechanisms, and skip connections, address specific challenges in image recognition tasks? 2

c) You are training a CNN on a small image dataset and notice overfitting. What architectural or regularization techniques can help reduce overfitting? 2

d) Explain the difference between feature maps and activation maps in the context of CNNs. 3